EN.625.740.81
Homework 6

1.

(a)

Show that the distance from the hyperplane $g(\mathbf{x}) = \mathbf{x^T x} + w_0 = 0$ to the point $\mathbf{x}_a$ is $|g(\mathbf{x}_a)|/\|\mathbf{w}\|$ by minimizing $\|\mathbf{x} - \mathbf{x}_a\|^2$ subject to the constraint $g(\mathbf{x}) = 0$.

(b)

Show that the projection of $\mathbf{x}_a$ onto the hyperplane is given by

$$\mathbf{x_p} = \mathbf{x}_a - \frac{g(\mathbf{x}_a)}{\|\mathbf{w}\|^2}\mathbf{w}$$

2.

Let $\mathbf{x_1}, \ldots, \mathbf{x_n}$ be n q-dimensional samples and Q be any nonsingular positive definite $q \times q$ matrix. Show that the vector $\mathbf{x}$ that minimizes

$$\sum_{k=1}^{n} (\mathbf{x_k} - \mathbf{x})^T Q^{-q} (\mathbf{x_k} - \mathbf{x})$$

is the sample mean, $\bar{\mathbf{x}} = \frac{1}{n} \sum_{k=1}^{n} \mathbf{x_k}$.

3.

Consider a linear classifier with discriminant functions $g_i(\mathbf{x}) = \mathbf{w_i^T x} + w_{i0}, i = 1, \ldots, c$. Show that the decision regions are convex by showing that if $\mathbf{x}_1 \in \mathcal{R}_i$ and $\mathbf{x}_2 \in \mathcal{R}_i$ then $\lambda \mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2 \in \mathcal{R}_i$ if $0 \leq \lambda \leq 1$.

4.

In the gradient descent algorithm $\mathbf{a}_{k+1}$ is obtained from $\mathbf{a}_k$ by

5.

Show that the partial derivatives of the functions $y_i = \exp(a_i)/\sum_k \exp(a_k)$ used in multiple class logistic discrimination are given by

$$\frac{\partial y_i}{\partial a_j} = y_i(\delta_{ij} - y_j)$$

where $\delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & \text{otherwise.} \end{cases}$

6.

In Module 1 we introduced nine (9) datasets. The data files are posted here in Module 9. Choose three (3) of these datasets and for each of the three:

(a)

i.

ii.

(a)

i.

ii.

iii.

iv.

A.